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Salivary cortisol, heart rate, electrodermal activity and subjective stress responses to the Mannheim Multicomponent Stress Test (MMST)

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ABSTRACT

The availability of effective laboratory paradigms for inducing psychological stress is an important requirement for experimental stress research. Reliable protocols are scarce, usually laborious and manpower-intensive. In order to develop an economical, easily applicable standardized stress protocol, we have recently tailored the Mannheim Multicomponent Stress Test (MMST). This test has been shown to induce relatively high stress responses without focusing on social-evaluative components. In this study we evaluated changes in electrodermal activity and salivary cortisol in response to the MMST. The MMST simultaneously combines cognitive (mental arithmetic), emotional (affective pictures), acoustic (white noise) and motivational stressors (loss of money). This study comprised two independent experiments. For experiment 1, 80 female subjects were recruited; 30 subjects (15 females) participated in experiment 2. Significant changes in electrodermal activity and salivary cortisol levels in response to MMST exposure were found. Subjective stress and heart rate responses were significantly increased in both experiments. These results indicate that the MMST is an economical stress paradigm which is also applicable in larger cohorts or multicenter studies for investigating stress reactions. As social-evaluative threat is not the main stress component of the MMST, this procedure represents a useful and complementary alternative to other established stress protocols.

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1. Introduction

Life exists by maintaining a complex dynamic equilibrium, or *homeostasis*, that is constantly challenged by intrinsic or extrinsic adverse forces or stressors (McEwen and Seeman, 2003; De Kloet et al., 2005). Stress can be considered as a state of threatened homeostasis and results in activating a complex repertoire of physiological and psychological responses. Acute psychobiological stress responses include the activation of the autonomic nervous system and the hypothalamo–pituitary–adrenal (HPA) axis, leading to changes in heart rate and blood pressure, as well as changes in adrenocorticotropin hormone (ACTH) and cortisol levels. On the subjective level stress responses include affective (e.g., fear, anger) as well as perceived physical changes (e.g., tension, trembling or sweating). An effective regulatory control of these processes triggered by stress is essential to health and survival (de Kloet et al., 2005).

Different studies have specifically focused on the effects of psychological stressors on stress reactions in a laboratory setting (Kirschbaum et al., 1993; Lejuez et al., 2003; Dickerson and Kemeny, 2004). Several stressor characteristics have been shown to influence stress responses. *Social-evaluative threat* occurs when a person is (potentially negatively) judged by others. Elements of the experimental protocol with social-evaluative threat include presence of evaluative others during the stress task or permanent recording of the performance for later evaluation. Stressors are defined as *uncontrollable* when it is difficult for participants to avoid negative consequences. Furthermore, stressors can be categorized as *motivated performance tasks* or *passive tasks*. Motivated performance stressors involve active performance situations that demand immediate overt or cognitive responses, whereas passive tasks, for example, include passive watching a film or being exposed to noise.

Laboratory studies established different acute standardized stressors that can be used for investigating stress reactions (Dickerson and Kemeny, 2004). The stressor tasks can be classified into five categories, namely cognitive, emotional, acoustic, public speaking and public speaking/cognitive task combination stressors. In a meta-analysis by Dickerson and Kemeny (2004) on acute psychological stress paradigms, social-evaluative threat and uncontrollability have been demonstrated to cause robust and reliable cortisol responses. In this review, stressors including a combination of public speaking and cognitive tasks were shown to be associated with greater cortisol responses. However, it should be noted that the stressors characterized by social-evaluative threat were not perceived as more distressing than other types of stressors, whereas autonomic changes were not included in this meta-analysis.

Different studies have shown that cognitive, emotional and acoustic stressors were associated with changes in physiological

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reactivity, neuroendocrine responses, and subjective stress experience (*cognitive stressors*: Condren et al., 2002; Mathias et al., 2004; Lejuez et al., 2003; *emotional stressors*: Meagher et al., 2001; Carter et al., 2008; *acoustic stressors*: Gerra et al., 1998). However, the effects of these stressors on the physiological system are highly variable. Dickerson and Kemeny (2004) have shown that there is substantial variability in the effect sizes depending on the nature of the stressor task. Some tasks provoked large cortisol responses (e.g. public speaking/cognitive task combinations: average Cohen's $d=0.87$) whereas others were associated with moderate or small effects (e.g. cognitive tasks: average Cohen's $d=0.20$). The best-established laboratory stress paradigm is currently the Trier Social Stress Test (TSST), which is composed of a brief preparation period followed by 5 min of public speaking (job interview) and 5 min of mental arithmetics in front of a panel of evaluators (Kirschbaum et al., 1992, 1993). The main reason for the success of this protocol is its capability to elicit robust neuroendocrine stress responses in the majority of subjects including increases in salivary cortisol and adrenocorticotropic hormone (ACTH) levels as well as in heart rate (Kudielka et al. 2004a, 2004b; Het et al., 2009). However, the TSST is relatively laborious and manpower-intensive. Furthermore, the availability of alternative complementary stress protocols with different psychological profiles would be useful for psychobiological research.

On this account, an economical stress paradigm has recently been developed in our group for inducing relatively high subjective stress without focusing on social-evaluative components (Kolotylova et al., 2010). We proposed that a combination and simultaneous presentation of established stressors focusing on different modalities would provide a context capable of triggering substantial stress-associated changes. This previous study comprised two independent experiments. In the first experiment involving 10 subjects, different cognitive stressors (the Paced Auditory Serial Addition Task (PASAT-C), 2-back test and Stroop test) were combined with emotional (a set of pictures with highly negative emotional valence), acoustic (white noise) and motivational (loss of money after each wrong answer in the cognitive task) stressors to investigate changes in heart rate and subjective stress. During each stress induction, all stressors of different modalities were used simultaneously. The duration of each stress session was 5 min, whereas the stress combinations were used in a permuted order. All stress combinations led to significant increases of heart rate (all p-values<0.01). At a descriptive level, a higher heart rate increase could be found during the stress induction with the Paced Auditory Serial Addition Task (PASAT-C, Lejuez et al., 2003) as a cognitive stressor. However, there were no significant differences in heart rate increases between the stress combinations. All stress combinations caused significant increases in subjective stress levels (all p-values<0.001). Following the stress induction with the PASAT-C, significantly higher ratings of subjective stress have been reported ($p<0.01$). Consequently, it could be assumed that this stress combination with the PASAT-C as a cognitive stressor, emotional, acoustic and motivational stressors represented a stronger subjective stressor. In the second experiment, this stress combination, the Mannheim Multicomponent Stress Test (MMST), was tested in an independent sample of 32 subjects. In this sample, we also found significant changes in subjective and objective stress measurements during the stress induction with the MMST (Kolotylova et al., 2010).

In laboratory studies, electrodermal responses are measured when it is desired to assess the differential or changing impact of stress (Andreassi, 2009). In different studies, the efficacy of the stress induction has been clearly demonstrated by the pronounced increase in electrodermal activity (Setz et al., 2010; Svetlak et al., 2010). Furthermore, salivary cortisol is frequently used as a biomarker of psychological stress (Hellhammer et al., 2009; Kudielka et al., 2009). Here, we present two studies that sought to replicate the previously observed changes in heart rate and subjective stress responses as well as to investigate electrodermal (study 1) and cortisol responses (study 2) to the MMST. A further aim of study 2 was to investigate sex differences in stress responses to the MMST.

2. Material and methods

2.1. Participants

Exclusion criteria in both studies comprised smoking, endocrine, cardiovascular or other severe medical diseases as well as schizophrenia, bipolar-I-disorder, acute depressive episode, substance dependence or abuse during the last 6 months. Psychiatric diagnoses were assessed by use of the Structured Clinical Interview for The Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) Axis-I Disorders and the SCID-II *Personality Questionnaire for DSM-IV Axis-II Disorders* (SCID I, First et al., 1997, SCID II, First et al., 1996). If the criteria from the SCID II Personality Questionnaire were met, the International Personality Disorder Examination (IPDE, Loranger, 1999) was additionally used. The participants were asked to avoid consumption of caffeine, extensive meal and comprehensive physical exercises 2 h before the beginning of the investigation. The study protocol was approved by the ethics committee of the University of Heidelberg and all participants provided written informed consent. For participation, they received monetary compensation (€40).

2.1.1. Study 1

Eighty female subjects were recruited via the residents' registration office in Mannheim, Germany, where the addresses of potential participants were randomly selected. They were on average 25.4 years old (standard deviation (S.D.)=4.5, range 18–35). As different studies have shown that there were no effects of menstrual phase on heart rate and electrodermal activity in response to psychological stress (Gómez-Amor et al., 1990; Stoney et al., 1990; Pico-Alfonso et al., 2007; Veldhuijzen van Zanten et al., 2009), we did not control for menstrual cycle phase in study 1.

2.1.2. Study 2

Thirty healthy subjects were recruited through flyers posted at the Universities of Mannheim and Heidelberg. The 15 female and 15 male participants were on average 24.7 years old (S.D.=4.6, range 18–38). Only females who did not use oral contraceptives and reported having a regular menstrual cycle between 28 and 35 days were included. The menstrual phase was calculated on the basis of the first day of last menses and the cycle length, which were both ascertained during the experimental session. All participants were in the luteal phase of their menstrual cycle at the time of examination (Kirschbaum et al., 1999; Kumsta et al., 2007). The luteal phase was defined as the last 14 days of the cycle.

2.2. Procedure

Participants arrived at the laboratory for individual afternoon sessions (starting between 13.00 h and 16.00 h, no stress exposure before 14.00 h). At first, written informed consent was obtained and the subjects received detailed instructions for saliva collection. Subsequently, subjects rested for 15 min in study 1 and for 20 min in study 2. During baseline conditions, subjects were alone in the room and were asked to relax. During the stress period, the participants solved the tasks in the presence of the investigator who was not in the participants' line of vision and did not give any comments. There was no pretend or real video- or tape-recording of the subjects' behavior. In both studies, heart rate (HR) was monitored continuously. Levels of subjective stress were determined after baseline conditions and immediately after induction of stress. In study 1, skin conductance level (SCL) and numbers of skin conductance fluctuations per min (NSCF) were measured during the whole investigation. In study 2, saliva samples were obtained following baseline conditions (T1), after stress induction (T2) as well as 10, 20, 30, and 40 min after cessation of stress (T3–T6).

2.2.1. Stress induction by the Mannheim Multicomponent Stress Test (MMST)

To economically elicit a relatively high level of stress, stressors of different modalities (e.g., cognitive, emotional, acoustic and motivational stressors) were used simultaneously and the stress induction took only 5 min.

As a *cognitive stressor*, we used a modified computer version of the Paced Auditory Serial Addition Task (PASAT-C, Lejuez et al., 2003). Numbers were sequentially flashed on a screen. Participants were asked to sum the most recent number with the previous number. Using a keyboard provided on the screen, they were instructed to click on the correct answer with the computer mouse. After providing each sum, the participant had to ignore the sum and add the following number to the most recently presented. In different studies, the PASAT-C has been used to assess interindividual responses to stress (Lejuez et al., 2003; Schmahl et al., 2010). For the MMST, level 1 and the first part of level 2 were used, both lasting 2 min. The time latency between the numbers was 3 s for level 1 and 2 s for level 2.

As an *emotional stressor*, a set of emotional photographic slides with different affective valence was applied. The set with negative affective valence (44 slides) included mainly disgust and fear related pictures, whereas the set with positive affective valence (nine slides) included pictures with babies and animals. For both sets, emotional scenes as well as facial expressions of emotions were used. The slides with negative valence were selected on the internet and from the International Affective Picture System (IAPS, Lang et al., 2005) whereas all slides with positive valence were taken from the IAPS. An independent sample of subjects rated the degree of valence, arousal and

specific emotions elicited by the slide shows (Kolotylova et al., 2010). It could be demonstrated that the slides with negative affective valence evoked unpleasant emotions (e.g., fear and disgust) and high arousal, whereas the slides with positive affective valence were rated as pleasant and evoked moderate arousal. After each block of five slides with negative valence, a slide with positive valence was presented to avoid habituation effects. The slides with negative valence were presented for five seconds, whereas the slides with positive valence were presented for three seconds. While performing the PASAT-C, the slides were demonstrated in the background. To induce a high level of attention towards the aversive emotional pictures, 13 slides were presented before the start of the PASAT-C. Three of these slides were demonstrated twice. The subjects were asked to verbally identify the slides that were presented twice.

As an *acoustic stressor*, we used white noise applied via headphone. The white noise was constantly applied during the stress induction. The intensity of the white noise was increased from 78 to 93 dB to avoid habituation (Flor and Grüsser, 1999).

To add a *motivational stressor*, the subjects were instructed that their participation reimbursement was reduced after each wrong answer. Starting with an amount of €100, an acoustic signal was presented for each wrong answer in the PASAT-C. However, every subject ended at €40 (compensation for study participation). Before leaving the lab, the subjects were debriefed by the investigator about the nature and purpose of the study.

The total duration of stress induction was 5 min. During the first minute, emotional pictures were shown and the white noise was induced. Thereafter, the cognitive tasks were started.

2.2.2. Apparatus and stress measures

Emotional and cognitive stimuli were presented on a computer screen. The subjects were seated 1.5 m in front of the screen. The presentation was computer-automated using the program "Presentation 0.71" (Neurobehavioral Systems, http://www.neurobs.com). In both studies, subjective stress levels were assessed using a Likert-scale ranging from 0 to 9 (0—no subjective stress experience at all; 9—extreme experience of subjective stress). In study 1, heart rate and electrodermal activity were registered by the ActiveTwo AD-box of the Biosemi system and collapsed over 5-min intervals. Two heart rate sensors were applied to the collarbone below the right shoulder and below the left breast, whereas two skin conductance sensors were attached to the inside of the right foot over the abductor hallucis muscle. The signals were continuously sampled at 2048 Hz and stored to a hard disk (Fig. 1).

In study 2, heart rates were measured continuously using wireless transmission with electrocardiogram precision at 5-s intervals and collapsed over 5-min intervals (Polar Instruments, Polar S 810i). To measure individual cortisol responses to stress exposure, saliva samples were obtained using the "Salivette" sampling device (Sarstedt, Nümbrecht, Germany). Samples were kept at room temperature throughout the test session and then stored at −20 °C until analysis. Before analysis, samples were centrifuged at 3000 rpm for 5 min. Salivary cortisol levels were determined employing a commercial immunoassay with chemiluminescence detection (CLIA, IBL; Hamburg, Germany). The lower detection limit of this assay is 0.43 nmol/l while intra- and inter-assay coefficients of variation are below 8% (Fig. 2).

2.3. Statistical analyses

Subjective stress ratings before and after the stress induction were analyzed by dependent *t*-test for within-group comparisons. Analyses of variance (ANOVAs) for repeated measures were computed to analyze heart rate, electrodermal activity as well as endocrine responses to the MMST. All stress parameters were approximately normally distributed (Kolmogorov–Smirnov-Test: all *p*-values > 0.30). When the Mauchly test of sphericity indicated covariance heterogeneity, repeated-measures were carried out using Greenhouse–Geisser corrections. In case of significant results, post hoc comparisons with Bonferroni correction were applied. In order to further illustrate significant individual prestress-peak-differences for subjective stress responses, heart rate and cortisol responses, we computed effect sizes *d* (Cohen, 1988). In both studies, all subjects exhibited peak heart rate and electrodermal response during the stress induction. For cortisol concentrations, individual peak responses could be observed at all poststress measurements (T2–T6).

P-values ≤ 0.05 (two-tailed) were considered statistically significant. All results shown are means ± S.D.

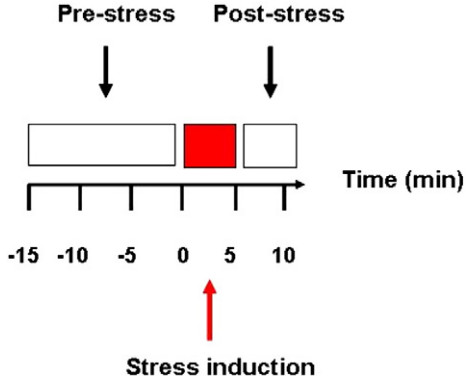


Fig. 1. Study 1. Experimental protocol.

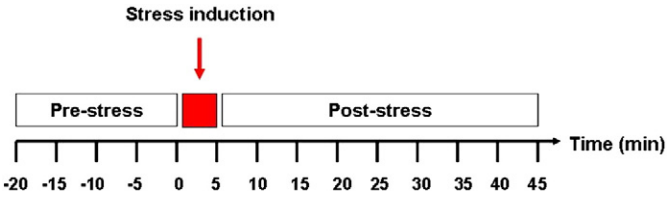


Fig. 2. Study 2. Experimental protocol.

3. Results

3.1. Study 1

Due to a technical problem, HR could not be assessed in three subjects, whereas SCL and NSCF were not assessed in two subjects.

3.1.1. Subjective stress ratings

As expected, a paired-samples *t*-test indicated a significant increase of subjective stress ratings in response to the MMST ($t = -26.354$, $p \leq 0.001$). From a baseline mean of 1.0 points (S.D. = 1.1, range 0.0–5.0), mean stress ratings reached 7.1 points after the stress induction (S.D. = 1.7, range 3.0–9.0). The individual prestress-peak differences ranged from 2 to 9 points. All subjects reported an increased subjective stress level after the stress induction with the MMST. The effect size for the individual prestress-peak differences reached 4.1, indicating a large effect.

3.1.2. Heart rate responses

In this sample, heart rate was significantly increased in response to the MMST exposure ($F_{(1.6;\ 122.8)} = 57.958$, $p \leq 0.001$). From a mean of 76.1 (S.D. = 12.3, range 56.2–112.5) bpm at the end of the baseline phase, the mean peak heart rate was 94.4 (S.D. = 16.2, range 62.6–128.9) bpm during the stress induction. The individual prestress-peak differences ranged between −48.2 bpm and 50.3 bpm. About 91% of the subjects showed a heart rate increase during the stress induction with the MMST. The effect size for the individual prestress-peak differences reached 1.3, indicating a large effect.

3.1.3. Electrodermal activity

Significant increases of SCL have been found in response to the stress induction ($F_{(1.3;\ 101.3)} = 52.950$, $p \leq 0.001$). From a mean of 5.8 (S.D. = 4.3, range 1.3–20.0) μS at the end of the baseline phase, the mean peak SCL was 9.3 (S.D. = 5.6, range 1.4–31.8) μS during the stress induction. The individual prestress-peak differences ranged between −10.8 μS and 12.8 μS (Fig. 4).

For NSFR, significant increases have been also found during stress induction with the MMST ($F_{(1.9;\ 149.5)} = 113.419$, $p \leq 0.001$). From a mean of 1.8 (S.D. = 2.4, range 0.0–9.8) fluctuations per min at the end of the baseline phase, the mean peak NSFR reached 6.9 (S.D. = 3.4, range 0.0–12.6) fluctuations per min during the stress induction. The individual prestress-peak differences ranged between −1.3 and 11.3 fluctuations per min.

About 95% of subjects showed an increase of SCL and NSFR during the stress induction with the MMST. The effect size for the individual prestress-peak differences reached 0.7 for SCL (a medium effect) and 1.7 for NSFR (a large effect).

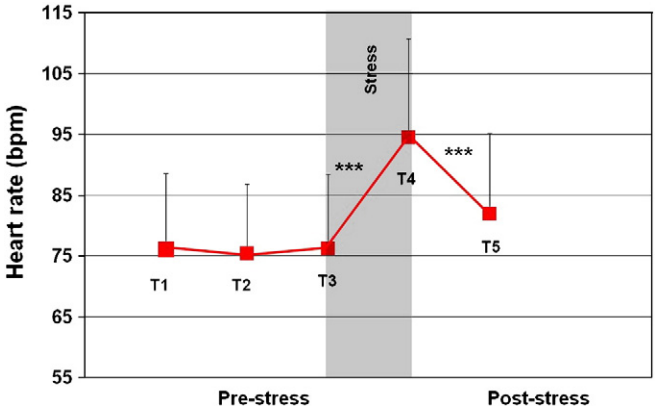


Fig. 3. Study 1 ($n=77$). Mean heart rate responses ($\pm$S.D.) to the Mannheim Multicomponent Stress Test (MMST). Heart rate was collapsed over 5-min intervals. T1, T2 and T3—baseline, T4—stress induction, T5—post-stress. The stars indicate significant differences between two measurements as assumed by post-hoc comparison tests with Bonferroni correction. $^{***}p \leq 0.001$.

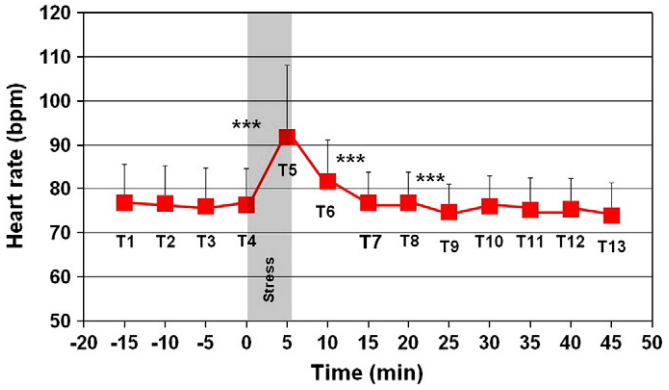


Fig. 5. Study 2 ($n=30$). Mean heart rate responses ($\pm$S.D.) to the Mannheim Multicomponent Stress Test (MMST). Heart rate sampled every 5 s was collapsed over 5-min intervals. −20–0 min—baseline, 0–5 min—stress induction. The stars indicate significant differences between two measurements as assumed by post-hoc comparison tests with Bonferroni correction. $^{***}p \leq 0.001$.

3.2. Study 2

3.2.1. Subjective stress ratings

After stress exposure, mean subjective stress ratings were significantly higher than before ($t=-9.766$, $p \leq 0.001$). From mean baseline ratings of 1.2 points (S.D. = 1.0, range 0–4), mean stress ratings reached 5.8 points after the stress induction (S.D. = 2.7, range 0–9). The individual prestress-peak differences ranged from 0 to 9 points. About 94% of the subjects reported an increased subjective stress level after the stress induction with the MMST. The effect size for the individual prestress-peak differences reached 1.9, indicating a large effect.

3.2.2. Heart rate responses

Significant increases in heart rate were observed in response to the stress induction ($F_{(1.96;\ 56.92)}=15.610$, $p \leq 0.001$). From a mean of 76.3 (S.D. = 8.3, range 60.3–91.8) bpm at the end of the baseline phase, the mean peak heart rate was 91.7 (S.D. = 16.4, range 64.9–124.3) bpm during the stress induction. Directly after stress induction, heart rate returned to baseline levels. The individual prestress-peak differences ranged between −4.9 bpm and 45.9 bpm. About 94% of the subjects showed a heart rate increase during the stress induction with the MMST. The effect size for the individual prestress-peak differences reached 1.1, indicating a large effect (Fig. 5).

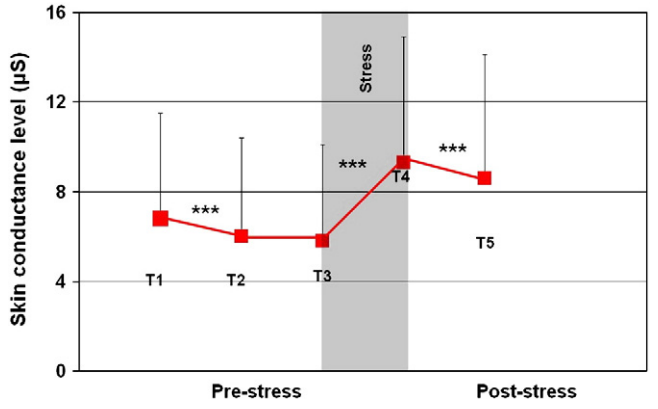


Fig. 4. Study 1 ($n=77$). Mean skin conductance level ($\pm$S.D.) to the Mannheim Multicomponent Stress Test (MMST). Skin conductance level was collapsed over 5-min intervals. T1, T2 and T3—baseline, T4—stress induction, T5—post-stress. The stars indicate significant differences between two measurements as assumed by post-hoc comparison tests with Bonferroni correction. $^{***}p \leq 0.001$.

3.2.3. Cortisol responses

As shown in Fig. 3, the participants showed a significant increase of salivary free cortisol in response to the MMST ($F_{(1.95;\ 56.50)}=7.344$, $p \leq 0.01$). As expected, a mean peak of cortisol level was observed 20 min after stress cessation. The effect size for the individual prestress-peak differences reached 0.6, indicating a medium effect (Fig. 6).

From pre-stress salivary cortisol concentrations ranging between 3.9 nmol/l and 21.5 nmol/l, individual peak levels after stress induction ranged from 3.7 nmol/l to 26.5 nmol/l. The individual prestress-peak differences ranged between −4.7 nmol/l and 13.6 nmol/l. 60% of the participants responded with an increase of cortisol above the individual pre-stress level. Ten percent of the participants responded with an increase of cortisol between 0 nmol/l and 1.4 nmol/l above individual pre-stress levels, whereas about 4% of the participants showed cortisol increases between 1.5 nmol/l and 2.4 nmol/l. About 47% of the participants showed cortisol increases of at least 2.5 nmol/l, which was proposed to be a conservative threshold for a significant cortisol increase (Weitzman et al., 1971; Wüst et al., 2000).

3.2.4. Sex comparison

Comparing sexes, ANOVA results revealed a trend for a main effect *Sex* ($F_{(1;\ 28)}=3.217$, $p=0.084$) with men showing higher mean cortisol prestress levels, while the subsequent net increases were almost

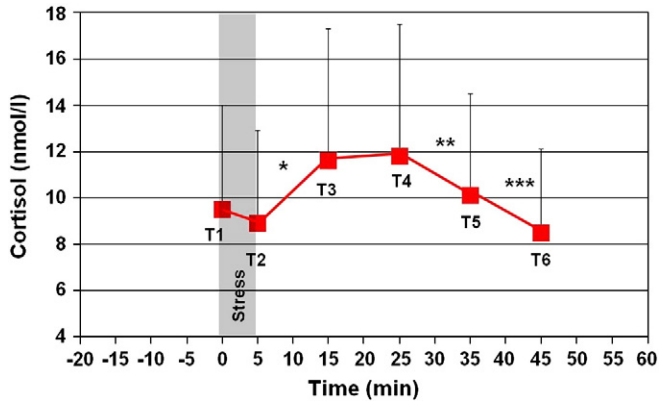


Fig. 6. Study 2 ($n=30$). Mean salivary cortisol responses ($\pm$S.D.) to the Mannheim Multicomponent Stress Test (MMST). −20–0 min—baseline, 0–5 min—stress induction. The stars indicate significant differences between two measurements as assumed by post-hoc comparison tests with Bonferroni correction. $^{*}p \leq 0.05$, $^{**}p \leq 0.01$.

identical in males and females (see Fig. 3). Regarding heart rate responses, neither the main effect for sex nor the interaction between timepoint and sex was significant (main effect *Sex*: $F_{(1;\ 28)}=0.211$, $p=0.65$; interaction *Sex*∗*Time*: $F_{(1;\ 28)}=0.408$, $p=0.66$). Likewise, subjective stress responses did not differ between males and females (main effect *Sex*: $F_{(1;\ 28)}=0.837$, $p=0.37$; interaction *Sex*∗*Time*: $F_{(1;\ 28)}=0.478$, $p=0.50$) (Fig. 7).

4. Discussion

In this study, we evaluated a recently developed economical and easily applicable stress paradigm for inducing relatively high subjective stress levels without focusing on social-evaluative components. As this stress protocol was developed in Mannheim (Germany) and consists of several components, it was named the "Mannheim Multicomponent Stress Test" (MMST). Since in previous studies only small or inconsistent effects with protocols employing one type of stressor tasks were observed, we combined different types of stressors to elicit significant and consistent stress responses. The MMST simultaneously combines cognitive, emotional, acoustic and motivational stressors and we demonstrated that it elicits stress responses on different levels. In two previous independent studies, the MMST protocol induced significant increases in subjective stress ratings as well as considerable changes in cardiovascular parameters (Kolotylova et al., 2010). In the present study, the effects of the MMST on subjective stress ratings, heart rate, electrodermal activity as well as HPA axis activity were investigated in two independent samples.

Subjective stress ratings and heart rate were significantly increased in response to the stress stimulation, indicating that the MMST has a reproducible and consistent effect on these measures. Furthermore, the MMST had a significant stimulatory effect on electrodermal activity. Regarding subjective stress ratings, Dickerson and Kemeny (2004) showed that all types of stressors (cognitive tasks, emotional tasks, public speaking/cognitive task combinations) elicited negative subjective responses (average Cohen's $d=1.0$). In our study, the effect size for subjective stress responses reached 4.1 in study 1 and 1.9 in study 2, indicating a large effect. Regarding heart rate and electrodermal activity, the effect sizes ranged in study 1 and study 2 between 0.7 and 1.7, indicating that the MMST has a stable effect on autonomic responses. In the meta-analyses of Dickerson and Kemeny (2004), autonomic responses have not been included. Therefore, direct comparisons of effects of the MMST and other stress paradigms on autonomic responses are impaired.

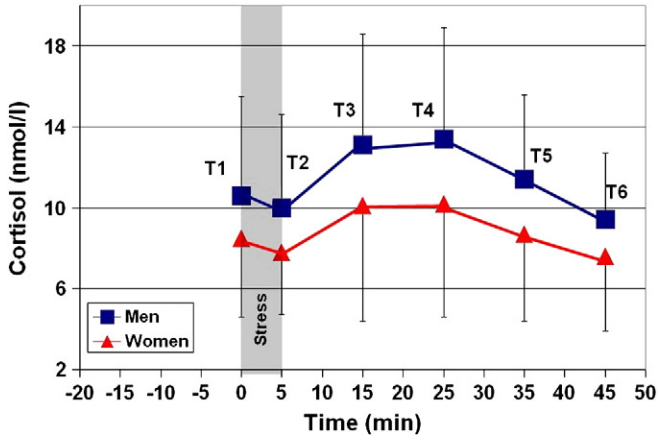


Fig. 7. Study 2 ($n=30$). Mean salivary cortisol responses (±S.D.) to the Mannheim Multicomponent Stress Test (MMST) in men and women in the luteal phase of the menstrual cycle. 0–20 min—baseline, 20–25 min—stress induction.

Moreover, the MMST elicited a significant cortisol response. The effect size for cortisol response reached 0.6, indicating a medium effect. Furthermore, cortisol responses to the MMST have been rather pronounced with responder rates of 60%, thus clearly exceeding cortisol increases that have been observed in response to single cognitive or emotional tasks (*cognitive tasks*: average Cohen's $d=0.20$; *emotional tasks*: average Cohen's $d=-0.13$, see Dickerson and Kemeny, 2004). It should be noted that stress paradigms combining cognitive tasks with a pronounced social-evaluative component (like the TSST) may elicit somewhat greater mean cortisol changes than the MMST (*public speaking/cognitive task combinations*: average Cohen's $d=0.87$, see Dickerson and Kemeny, 2004). However, the fact that the MMST is less laborious and manpower-intensive than the TSST and the explicit abandonment of a social-evaluative component may represent attractive features for many research projects thus outweighing the smaller mean cortisol increases. Furthermore, the availability of alternative complementary stress protocols with different psychological profiles would be useful for psychobiological research. In this study, 47% of the subjects showed cortisol increases of 2.5 nmol/l that has been viewed as a secretory episode (Weitzman et al., 1971; Wüst et al., 2000). On the other hand, Wüst et al. (2009) discussed that differences in the HPA stress responses between certain groups can perhaps be better detected with milder stressors even at the cost of a relatively high proportion of cortisol non-responders.

Regarding sex differences, the MMST elicited consistent effects on salivary cortisol, heart rate and subjective stress levels in females and males. Descriptively, men showed higher saliva free cortisol concentrations than women. However, only a trend could be found for a main effect for sex. These results are consistent with previous studies, which reported comparable saliva cortisol stress responses in men and women in the luteal phase of the menstrual cycle as response to stress induction, whereas descriptive analyses showed higher saliva cortisol levels in men (Kirschbaum et al., 1999; Rohleder et al., 2001; Wolf et al., 2001). In many of these studies, the TSST was used as psychosocial stress paradigm and it has been hypothesized that the well documented sex differences in cortisol stress responses can partly be ascribed to the specific psychological profile of this paradigm. However, in our study, we did observe very similar sex effects using a stressor with a differing psychological profile. Although we cannot rule out that sex-specific stress scenarios are conceivable, our findings do not support the assumption that the nature of the stressor has a major impact on cortisol response differences between males and females. Remarkably, males and females showed very similar subjective stress responses in our study, while for the TSST higher stress ratings for females have been observed (Kudielka et al., 1998). On a descriptive level, women showed larger mean increases in heart rate in response to stress but this difference did not reach statistical significance.

A limitation of our study is that we relied on self-reported menstrual cycle phases that were based on a single item. Furthermore, we didn't assess emotional states (e.g. anxiety, anger) after stress induction with the MMST. These emotional states may be associated with stress responses. In future studies, it would be important to evaluate the psychological responses to the MMST stimulation, using established psychological questionnaires. It should be noted that the MMST presents a combination of multiple components (cognitive, emotional, acoustic and motivational stressors) all of which can contribute to the observed physiological responses. Different groups might respond to different elements within this paradigm. However, the MMST has been developed for eliciting significant stress responses in the majority of subjects tested.

In this study, the presence of the investigator in the room during the stress induction, even if not in the participant's line of vision and not evaluative, may have created a minor social-evaluative context contributing to the observed cortisol responses. However, a study by Dickerson et al. (2008) showed that the mere social

presence of others did not drive the changes in cortisol responses. For this reason, we do not believe that a substantial social-evaluative context led to the observed cortisol changes.

Taken together, the present study suggests that the MMST is an economical stress paradigm that can be used for investigating stress reactions in a laboratory setting, without larger resources in terms of space, presence of an evaluative audience or intensive training of investigators. Furthermore, this stress paradigm can be used in larger cohorts or multicenter-studies even under field conditions. Compared to other stress paradigms employing only one type of stress task (e.g. cognitive or emotional stressors), the MMST was associated with greater cortisol and subjective stress responses. Unlike the established public speaking/cognitive task combinations like the TSST, social-evaluative threat is not the main stress component of the MMST. Therefore, this procedure represents a useful complementary alternative to these stress protocols. In future studies, it would be important to evaluate physiological and psychological responses to the MMST stimulation in healthy and clinical populations as well as cortisol responses in different menstrual cycle phases.

Disclosure

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper. All authors contributed to and have approved the final manuscript.

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